# Prefill vs Decode

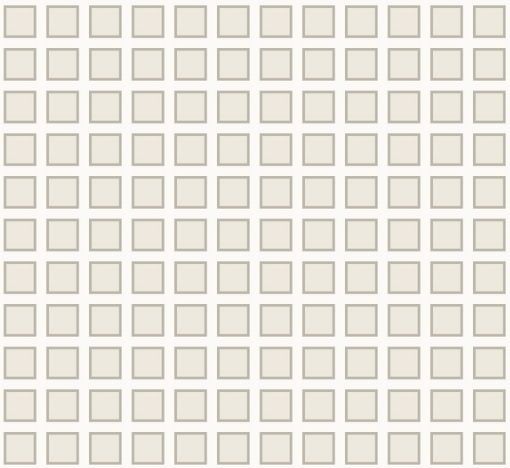
## same 64 layers, two very different jobs for the GPU

### Prefill (prompt phase)

input: 512 prompt tokens



weights in GPU memory: 51.2 GB

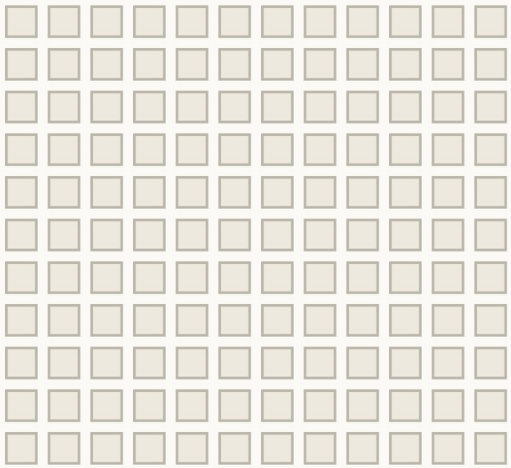


### Decode (one token at a time)

input: 1 new token

token 1

weights in GPU memory: 51.2 GB



## Prefill vs decode

### same 64 layers, same kernels, batch 1

Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency

# Prefill vs Decode

## same 64 layers, two very different jobs for the GPU

### Prefill (prompt phase)

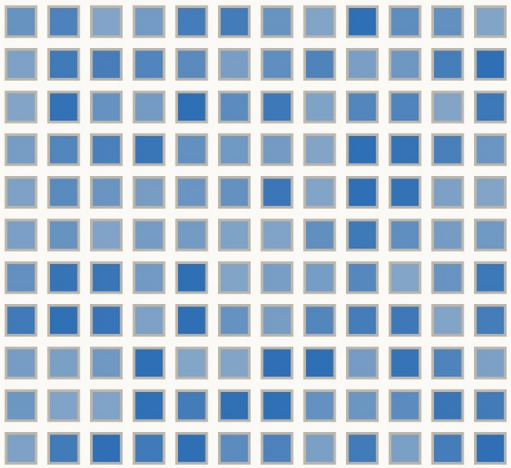
input: 512 prompt tokens



weights in GPU memory: 51.2 GB



Tensor Cores busy



t = 26.4 ms · 512 tokens in flight

### Decode (one token at a time)

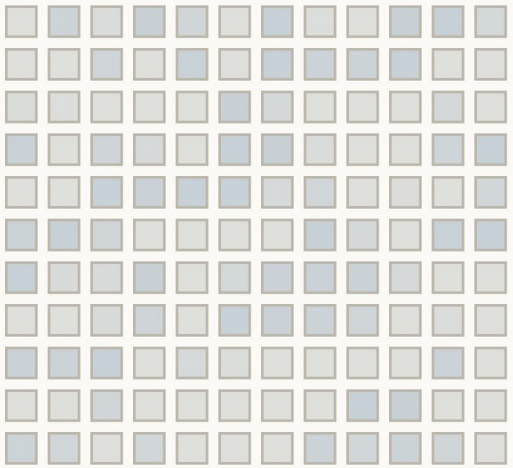
input: 1 new token

token 2

weights in GPU memory: 51.2 GB



SMs mostly waiting for weights



t = 26.4 ms · 1 token done

## Slow motion: the first 54 ms

prefill: 512 tokens at once · decode: 1 at a time



Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency

# Prefill vs Decode

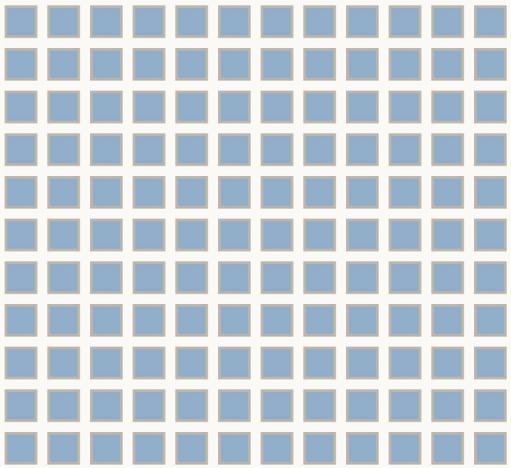
## same 64 layers, two very different jobs for the GPU

### Prefill (prompt phase)

input: 512 prompt tokens



weights in GPU memory: 51.2 GB



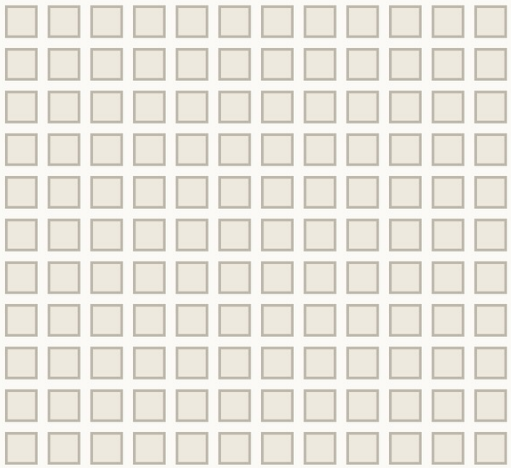
t = 54.3 ms · 512 tokens done

### Decode (one token at a time)

input: 1 new token

token 3

weights in GPU memory: 51.2 GB



t = 54.3 ms · 2 tokens done

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## Decode re-reads all 51.2 GB per token

prefill reads them once for 512 tokens

Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency

# Prefill vs Decode

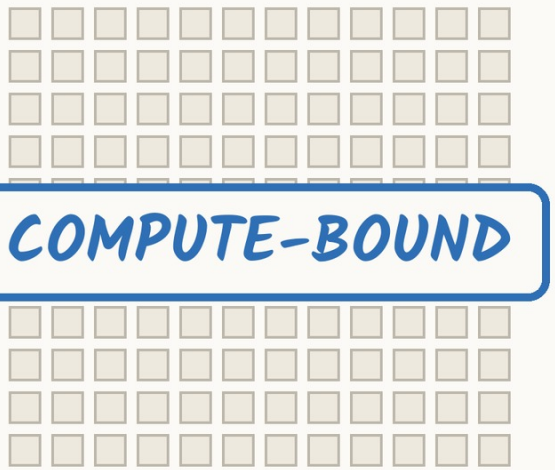
## same 64 layers, two very different jobs for the GPU

### Prefill (prompt phase)

input: 512 prompt tokens



weights in GPU memory: 51.2 GB




t = 54.3 ms · 512 tokens done

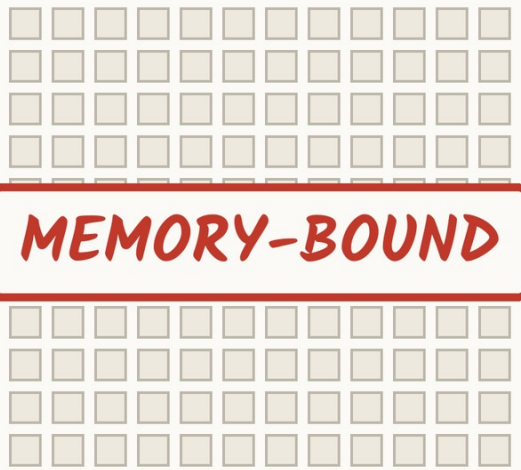
| | | |
|---|---|---|
| compute | 459 TFLOP/s (46%) | |
| memory | 0.94 TB/s (28%) | |

### Decode (one token at a time)

input: 1 new token

token 3

weights in GPU memory: 51.2 GB




t = 54.3 ms · 2 tokens done

| | | |
|---|---|---|
| compute | 2.49 TFLOP/s (0.25%) | |
| memory | 2.49 TB/s (74%) | |

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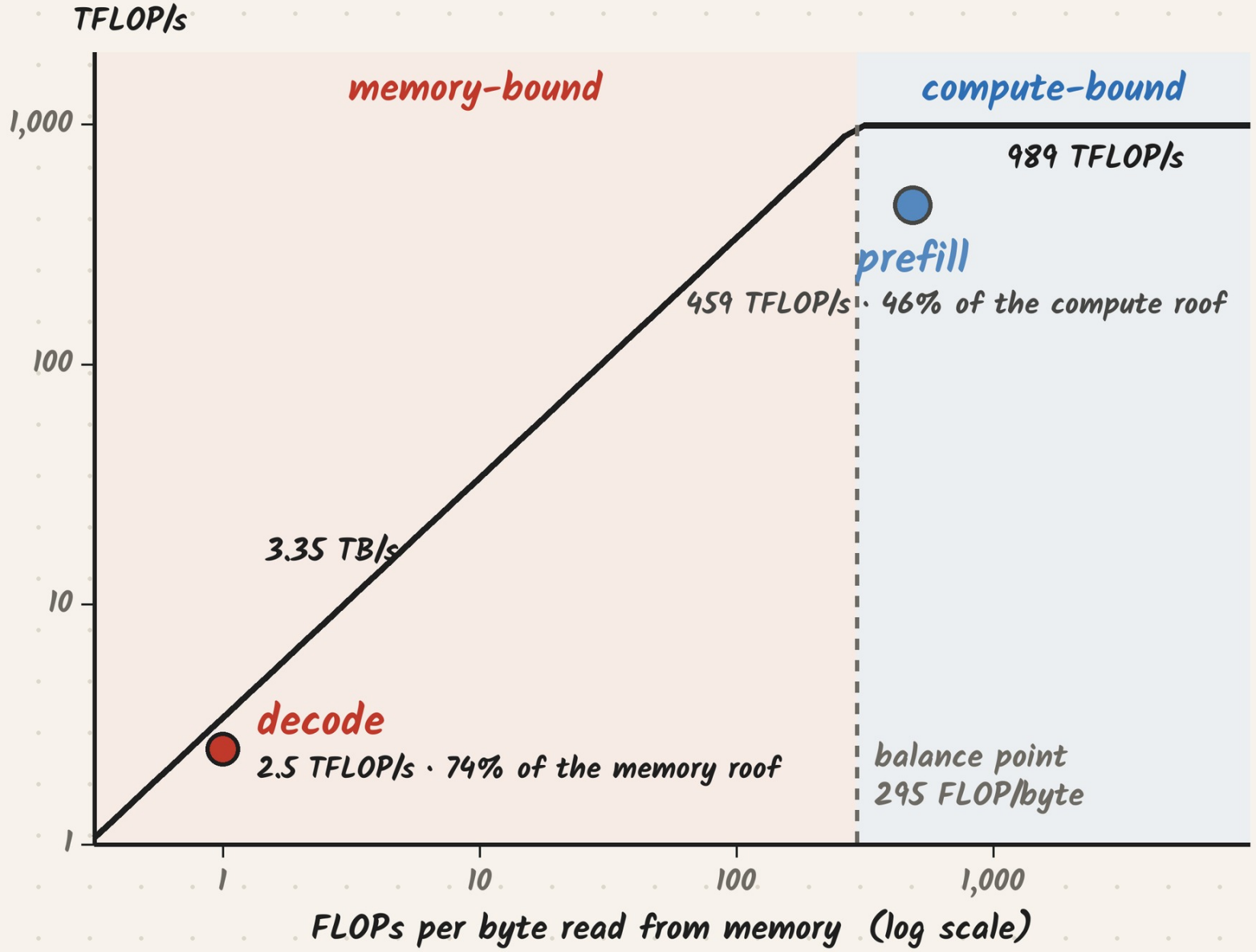
## Prefill: compute-bound

### decode: memory-bound, waiting for weights

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# Prefill vs Decode

## same 64 layers, two very different jobs for the GPU




S17

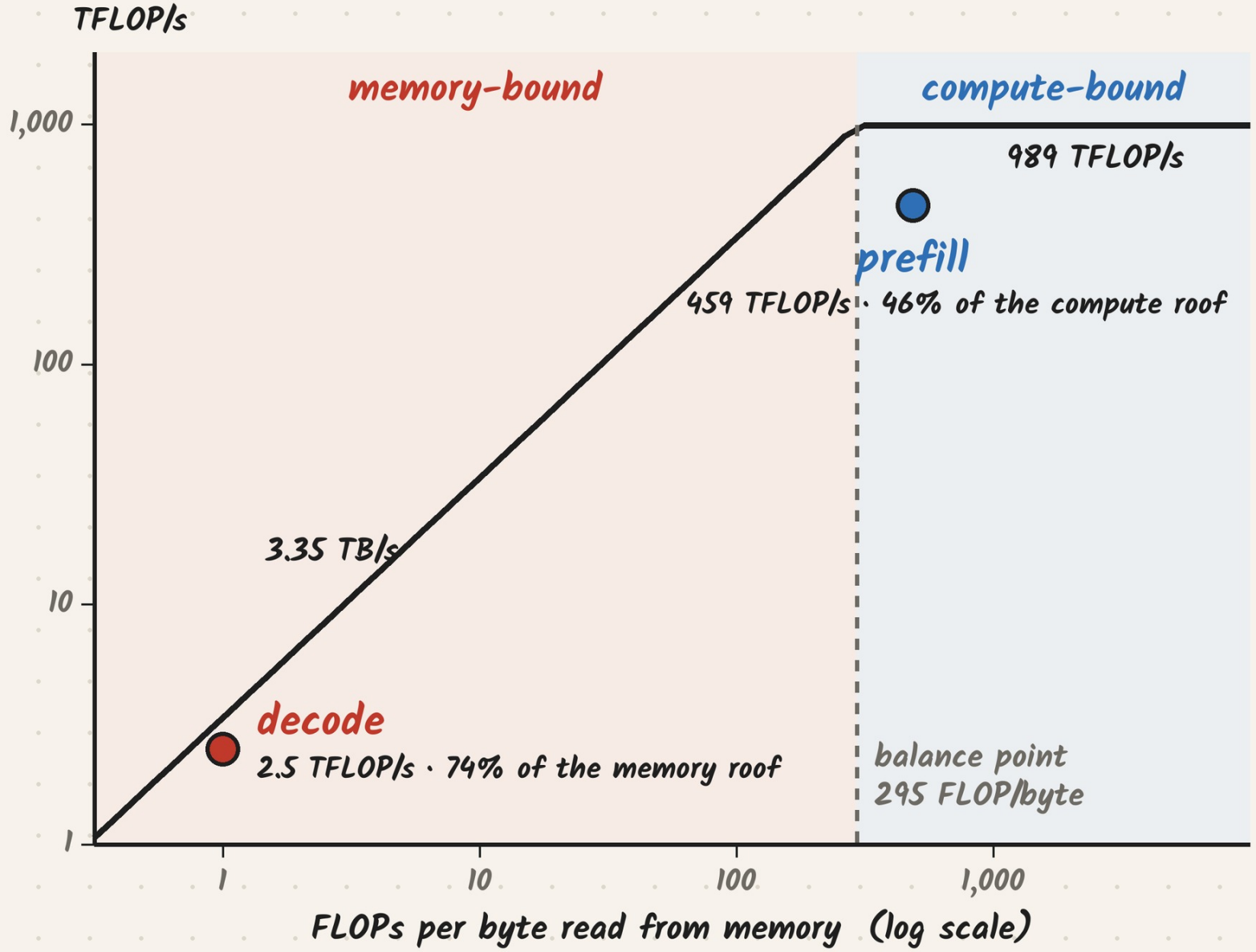
## The roofline: what limits a kernel

under 295 FLOP/byte: memory, above it: compute

Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency

# Prefill vs Decode

## same 64 layers, two very different jobs for the GPU




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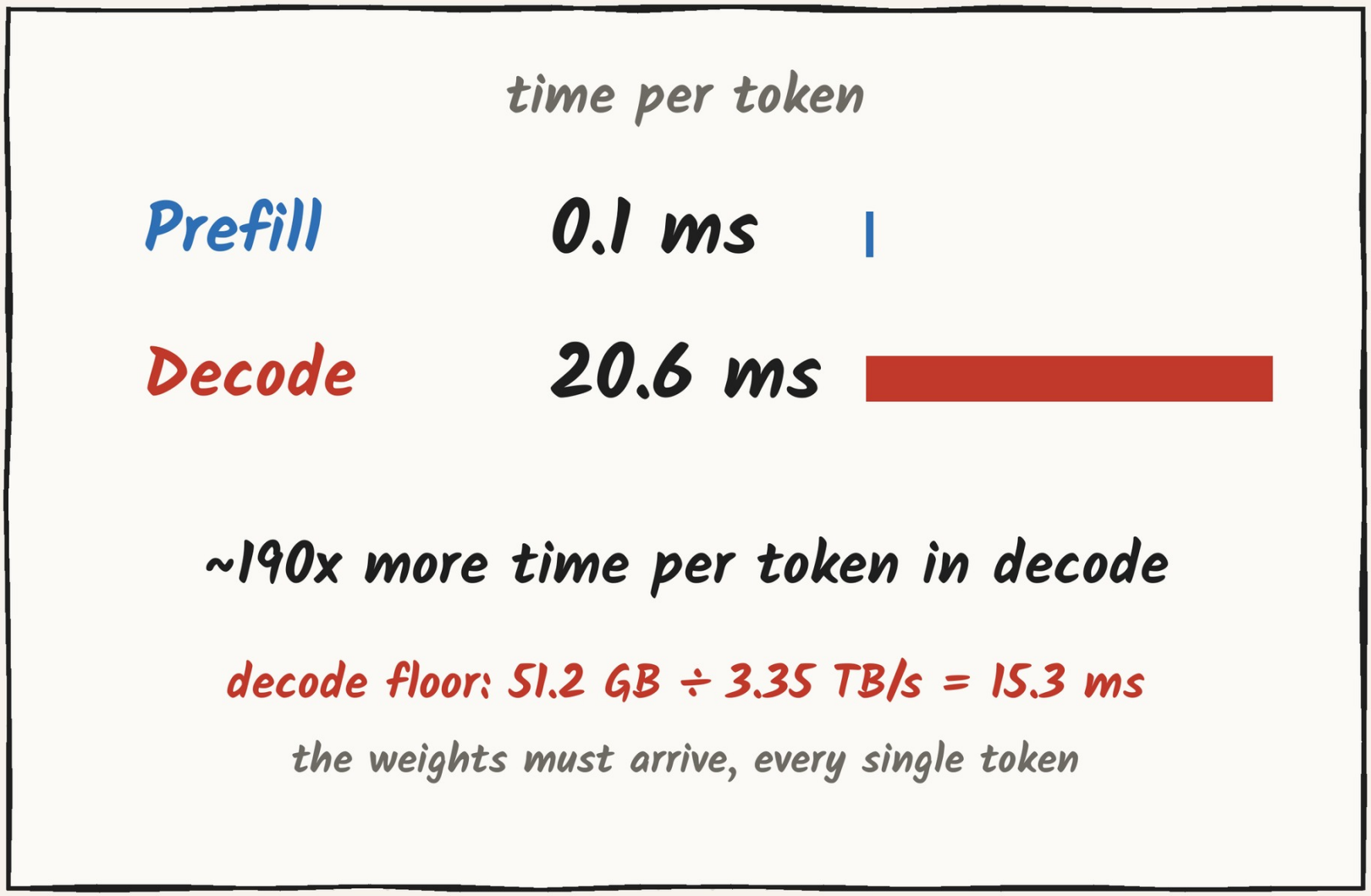
## Prefill ~490 vs decode ~1 FLOP/byte

each weight byte: 512 tokens vs 1 token

Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency

# Prefill vs Decode

same 64 layers, two very different jobs for the GPU




~190x more time per token in decode

decode floor: 51.2 GB ÷ 3.35 TB/s = 15.3 ms

the weights must arrive, every single token

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## Decode uses 0.25% of the compute

next: 32 users share one read of the weights

Qwen3.8-27B BF16 · vLLM 0.31 · 1x H100 SXM · batch 1 · 512-token prompt · vllm bench latency